Invariants, monovariants and extremal arguments

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L4-CO1 Level 4 Olympiad · Combinatorics

How to prove that a process can never reach a state, that the process must stop, or that a configuration has a property, by finding the right quantity or the right extreme object.
By the end you can find invariants (parity, residues, products, colourings, weighted sums) and use them to prove impossibility; prove that a process terminates using a bounded integer monovariant, and pair the monovariant with a strategy to find a maximum number of moves; choose an extreme object and turn what the extreme forces into a complete olympiad proof.
Prerequisites: Modular arithmetic (L3-NT1), Stars and bars, inclusion-exclusion and bijections (L3-CO1). **Contests:** BMO1, MOG, AIME, USAJMO. **Time:** about 120 minutes.

1 Lesson

Three frogs sit on the number line at 0, 1 and 3. A move consists of one frog jumping over another frog and landing the same distance beyond that frog, so a frog at x jumping over a frog at y lands at $2y - x$. Can the frogs ever sit at 10, 11 and 12?

Experimenting will not settle this, since the frogs can wander arbitrarily far. Consider a single frog instead. This frog goes from x to $2y - x$, and $2y - x$ has the same parity as x. So each frog keeps its parity for ever. At the start two frogs are on odd numbers and one is on an even number, but at 10, 11, 12 two would be on even numbers. So the frogs can never sit at 10, 11 and 12.

The same method applies whenever a problem describes a process (moves, operations, a game, a sequence of choices). Instead of following every possible history, find a quantity whose change under an arbitrary move you can compute, and argue from that quantity alone.

1.1 Invariants

Invariant principle.
Suppose a process moves between states, and I is a quantity computed from the state with $I(\text{after}) = I(\text{before})$ for every legal move. Then every state reachable from the starting state S_0 has $I = I(S_0)$. In particular, if $I(T) \neq I(S_0)$, the state T cannot be reached.

The proof is induction on the number of moves, and nobody will ask you for more than that sentence. The difficulty lies in finding I. The usual candidates, roughly in the order to try them:

- parity, or a residue modulo some m (work out what the move does mod 2, 3, 4, and so on).
- a sum, a product, or a sum of squares of the numbers involved.
- a colouring, where the positions are labelled so that each move touches the colours in a fixed pattern.

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- a weighted sum $\sum w_i c_i$, with weights chosen so that the effect of a move cancels.

Two habits matter. First, compute what the move does to a candidate quantity in general, not on one example. Second, remember that an invariant only ever proves that something is *impossible*. If the answer is "yes, the state can be reached", you must show how.

Example 1.
A machine holds a pair of positive integers, starting with $(1,2)$. In one step the machine replaces (a,b) by either $(a+2b,\, b)$ or $(a,\, b+2a)$. Can the machine ever hold $(60,210)$? What about $(2025,2026)$? And $(13,2)$?

Solution.
The sum increases at every step and gives no information, so we look at residues. The first kind of step changes a by $2b$, which is even, and leaves b alone, and the second changes b by $2a$ and leaves a alone. So a is always odd and b is always even. That rules out $(60,210)$ at once.
The pair $(2025,2026)$ passes this test. The pair also passes sharper ones. Note that since b is even, $2b$ is a multiple of 4, so in fact $a \equiv 1 \pmod 4$ for ever, and $2025 \equiv 1 \pmod 4$. Also $\gcd(a+2b,b) = \gcd(a,b)$, so the gcd stays equal to 1, which is true of 2025 and 2026. None of these invariants rules out $(2025,2026)$, so we look instead at the relative sizes of the entries after a step.
After any step, the entry that was just changed is more than twice the other one, because $a+2b > 2b$ and $b+2a > 2a$. So every pair the machine reaches, apart from the starting pair, has one entry more than double the other. But $2025 < 2\cdot 2026$ and $2026 < 2\cdot 2025$, so $(2025,2026)$ is unreachable.
(Working backwards.) The same observation decides any target, by running the machine backwards. The last step must have increased the larger entry, so the previous pair is forced. From $(13,2)$ we go back to $(9,2)$, then $(5,2)$, then $(1,2)$, so $(13,2)$ is reachable in three steps.

In this example the invariants gave only necessary conditions, and the impossibility of $(2025,2026)$ needed a separate argument about the last move. Many contest problems need both kinds of argument.

Example 2.
The hundred numbers $\frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \ldots, \frac{1}{202}$ are written on a board. A move consists of erasing two numbers a and b and writing $a+b-2ab$. After 99 moves one number is left. Show that this number does not depend on the order of the moves, and find the number.

Solution.
The statement implies that something is invariant, but the sum is not, because of the $-2ab$. When an operation has the shape "sum minus a multiple of the product", try to factorise. Here

$$(1-2a)(1-2b) = 1-2a-2b+4ab = 1-2(a+b-2ab).$$

So if we replace each number x by $1-2x$, the move becomes the erasure of two numbers and the writing of their product.
Hence $P = \prod(1-2x)$, taken over the numbers on the board, is invariant. For $x = \frac{1}{2k+2}$ we get $1-2x = \frac{k}{k+1}$, so at the start

$$P = \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4}\cdots\frac{100}{101} = \frac{1}{101}.$$

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If N is the final number then $1 - 2N = \frac{1}{101}$, so $N = \frac{50}{101}$, whatever the order of the moves.

1.2 Monovariants

Sometimes nothing is preserved, but something only ever moves one way. Typical questions are "prove that the process must stop", "prove that eventually ...".

Monovariant principle.
If V is a non-negative integer and every move decreases V by at least 1, then the process stops after at most $V(S_0)$ moves. Likewise, if V is an integer that increases by at least 1 with every move and never exceeds a fixed number B, then at most $B - V(S_0)$ moves are possible.

Proof. After k moves, $V \leq V(S_0) - k$, and $V \geq 0$, so $k \leq V(S_0)$. The integrality matters. A real quantity can decrease for ever (think of $1, \frac{1}{2}, \frac{1}{3}, \ldots$), so if your V is not an integer you need another reason why V cannot keep decreasing, for instance that there are only finitely many states and none can repeat.

Example 3.
Finitely many integers are written on a board. In one move, choose two of them, a and b, with $a \geq b + 3$, erase them and write $a - 1$ and $b + 2$. Prove that, however the moves are chosen, the process must stop.

Solution.
The move brings two numbers closer together, which suggests the sum of squares as a measure of spread. With $a = 10$, $b = 0$ the sum of squares goes from 100 to $81 + 4 = 85$, but with $a = 4$, $b = 1$ the sum of squares goes from 17 to $9 + 9 = 18$, so the sum of squares is not a monovariant.
The sum goes up by exactly 1, since $(a - 1) + (b + 2) = a + b + 1$. An increasing integer quantity is enough if the quantity is bounded above, so we look for a bound. Let M be the largest number at the start. In a move, $a - 1 < a \leq M$ and $b + 2 \leq a - 1 < M$, so the largest number on the board never increases, and every number stays at most M.
If there are n numbers (the move does not change how many), the sum is therefore always at most nM. The sum starts at some integer S_0 and goes up by 1 with each move, so at most $nM - S_0$ moves can be made. The process stops.

Often the monovariant is a simple quantity such as the sum, and the real work is a bound for the sum from a separate observation (here, that the largest number never increases).

1.3 The extremal principle

Extremal principle.
Every non-empty finite set of real numbers has a largest and a smallest element, and every non-empty set of positive integers has a smallest element. So in any finite configuration you may pick an extreme object (largest number, shortest cycle, closest pair, leftmost point) and ask what its extremeness forces.

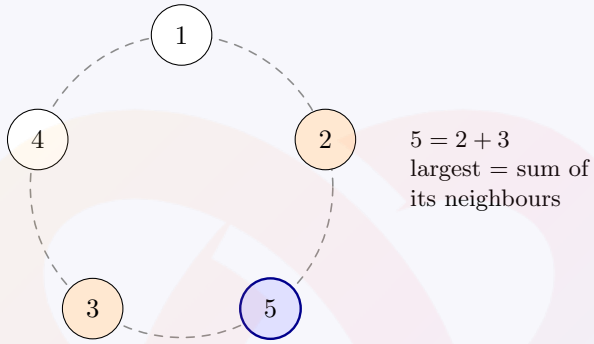
This is the same idea as induction and infinite descent, since "take a smallest counterexample" is

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an extremal argument. The difficulty is choosing which extreme to look at, and if one choice gives nothing, try another.

Example 4.
At least three distinct positive integers are arranged around a circle so that each one divides the sum of its two neighbours. Prove that the smallest number divides every number on the circle.

Solution.
First get examples. With three numbers $x < y < z$ we see that z divides $x + y$, which is positive and less than $2z$, so $x + y = z$. Then y divides $x + z = 2x + y$, so y divides $2x$, which is less than $2y$, giving $y = 2x$. So three numbers must be $x, 2x, 3x$. A bigger example is $1, 2, 5, 3, 4$ in that order around the circle, where $1 \mid 6$, $2 \mid 6$, $5 \mid 5$, $3 \mid 9$ and $4 \mid 4$.



In that example the largest number, 5, is the sum of its neighbours, which suggests looking at the largest number. Let M be the largest number and a, b its neighbours. They are smaller than M, so $0 < a + b < 2M$, and $M \mid a + b$ forces $a + b = M$.
Now suppose there are at least four numbers, and remove M, so that a and b become neighbours. We check that the property survives. The other neighbour of a is some c, and $c \neq b$ because there are at least four numbers. Before, $a \mid M + c$, and now we need $a \mid b + c$, and this holds since $b + c = (M + c) - a$. The same works for b, and nobody else's neighbours have changed.
(Removing the extreme element.) Hence we induct on the number n of integers. For $n = 3$ they are $x, 2x, 3x$ and the smallest, x, divides all of them. For $n \geq 4$, remove the largest number $M = a + b$. The remaining $n - 1$ numbers are distinct and still have the property, so by induction their smallest, m, divides all of them, in particular a and b, and hence $m \mid M$. Since M was the largest, m is also the smallest of all n numbers. This completes the induction.

Common traps.

- Using an invariant to "prove" that something is possible. Invariants only rule things out, so a "yes" needs a construction.
- Checking a quantity on one or two sample moves and declaring the quantity invariant. Write down the change for a general move, and check every type of move, including moves at the ends of a row or across the join of a circle.
- A monovariant that is not an integer, or is not bounded. Decreasing is not enough on its own.

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- Taking "the largest" element of a set that might be infinite. The extremal principle needs the extreme object to exist.
- In a write-up, forgetting to state the invariant clearly, compute the invariant at the start and at the target, and say why the two values differ.

1.4 How to spot this in a contest

The words "a move consists of", "repeatedly", "can we reach", "whatever choices are made" or "prove that the process terminates" indicate a process. Look for an invariant when the question is about reaching a state, and a monovariant when the question is about stopping. Start by doing a few moves by hand and writing down what changes and what does not, then test parity, residues, sums, products and colourings, in roughly that order.

When the problem is a fixed configuration and asks you to prove that something exists, or that every element has some property, try the extremal principle and take the largest, smallest, shortest or closest object and see what that object forces. If a problem asks for the largest possible number of moves, you usually need both halves, namely a monovariant for the upper bound and a careful strategy that achieves the bound.

2 Practice problems

The problems are roughly in order of difficulty, from $\star$ to $\star\star\star\star$. Hints are in Section 4 and full solutions in Section 5.

Problem 1 $\star$ The numbers $1, 2, 3, \ldots, 10$ are written on a board. In one move, choose two numbers a and b, erase them, and write $\frac{ab}{a+b}$. After nine moves a single number remains. Find this number.

Problem 2 $\star\star$ Twelve lamps stand in a circle, all switched off. In one move, choose three consecutive lamps and switch each of them (on becomes off and off becomes on). How many of the $2^{12} = 4096$ on/off patterns can be reached?

Problem 3 $\star\star$ A word is made of the letters A and B. A move replaces some occurrence of AB (an A immediately followed by a B) by BAA. Prove that, starting from any word, one cannot make moves for ever, and that the number of moves made before no move is possible does not depend on the choices.

Problem 4 $\star\star\star$ Ten coins lie in a row, all showing heads. In one move, choose a coin showing heads, turn that coin over, and also turn over the coin immediately to its right, if there is one. Prove that the moves must eventually stop, and find the largest possible number of moves.

Problem 5 $\star\star\star$ In a chess tournament with $n \geq 3$ players, every two players played each other exactly once and no game was drawn. Every player won at least one game. Prove that there are three players A, B, C such that A beat B, B beat C and C beat A.

Problem 6 $\star\star\star\star$ Six bowls stand in a circle. One of them, bowl A, contains a single bean, and the others are empty. In one move, take a bean out of any bowl and put one bean into each of the

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two neighbouring bowls. Prove that the bowl opposite A can never contain more beans than the other five bowls together.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **AMC 10B 2013 Problem 17** ⋆
   Give each colour of token a value so that every trade keeps the total value fixed.
2. **AMC 10B 2024 Problem 16** ⋆⋆
   Track the count of numbers modulo 3, and ask whether the number of odd numbers can ever go up.
3. **BMO Round 1 2011/12 Problem 4** ⋆⋆
   For each version, either find a strategy or find a quantity the moves cannot change.
4. **BMO Round 1 2023/24 Problem 2** ⋆⋆
   Look at how the difference between consecutive terms can change from one step to the next.
5. **AIME I 2012 Problem 11** ⋆⋆⋆
   Find a combination of the coordinates that every jump leaves unchanged mod 3, then decide which such points are reachable.
6. **USAJMO 2013 Problem 2** ⋆⋆⋆
   Start from a cell holding the smallest number.
7. **USAJMO 2020 Problem 1** ⋆⋆⋆
   Can the same two books ever be swapped twice?
8. **BMO Round 1 2019/20 Problem 5** ⋆⋆⋆
   Colour the seats alternately and see what a move does to the difference between the two colours.
9. **USAJMO 2019 Problem 1** ⋆⋆⋆⋆
   Look at the fruit in odd-numbered bowls and what one move does to this number.
10. **USAJMO 2015 Problem 6** ⋆⋆⋆⋆
    Ask what a move preserves in every row and every column.
11. **BMO Round 1 2025/26 Problem 6** ⋆⋆⋆⋆
    Number the pads modulo 4 and track how many frogs sit in each class.

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4 Hints

1. What happens to the reciprocal of the new number?
2. Label the lamps 0 to 11 and group them by their label modulo 3. How many lamps of each group does one move switch? For the construction, see what two overlapping moves do together.
3. Give each A the weight 2^k, where k is the number of Bs to its right, and add up the weights.
4. Give the coins weights $10, 9, \ldots, 1$ from left to right and add up the weights of the coins showing heads. For the construction, which head should you choose so that this total drops as slowly as possible?
5. First show that there is a cycle of players, each of whom beat the next, the last having beaten the first. Then take the shortest such cycle.
6. Look for weights $w_0, \ldots, w_5$ on the bowls such that a move never changes $\sum w_i c_i$. A move at bowl k removes w_k and adds $w_{k-1} + w_{k+1}$.

5 Solutions

Problem 1

The numbers $1, 2, 3, \ldots, 10$ are written on a board. In one move, choose two numbers a and b, erase them, and write $\frac{ab}{a+b}$. After nine moves a single number remains. Find this number.

Since $\frac{a+b}{ab} = \frac{1}{a} + \frac{1}{b}$, the reciprocal of the new number is the sum of the reciprocals of the two erased numbers. So the sum of the reciprocals of the numbers on the board is invariant. At the start this sum is

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{10} = \frac{7381}{2520}.$$

At the end this sum is $\frac{1}{N}$, where N is the last number. Hence $N = \frac{2520}{7381}$, whatever moves are made.

Answer: 2520/7381

Problem 2

Twelve lamps stand in a circle, all switched off. In one move, choose three consecutive lamps and switch each of them (on becomes off and off becomes on). How many of the $2^{12} = 4096$ on/off patterns can be reached?

Label the lamps $0, 1, \ldots, 11$ around the circle and put lamp i in class i mod 3, so each class has four lamps. Three consecutive labels have three different residues modulo 3, and this is still true across the join (for example $11, 0, 1$ have residues $2, 0, 1$) because 12 is a multiple of 3. So every move switches exactly one lamp in each class, and the parities of the numbers of lit lamps in the three classes all change together. At the start all three parities are even, so in every reachable pattern the three classes have lit-lamp counts of the same parity.

The number of patterns with this property is $2 \cdot 8^3 = 1024$, since we choose the common parity, and then each class of four lamps has 8 subsets of that parity.

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Conversely, all of these can be reached. The moves on lamps $i, i+1, i+2$ and on $i+1, i+2, i+3$ together switch only lamps i and $i+3$. Doing this for i and then for $i+3$ switches lamps i and $i+6$, and so on, so we can switch any two lamps of the same class and leave everything else unchanged. If a target pattern has an even number of lit lamps in every class, pair up its lit lamps within each class and switch the pairs. If the pattern has an odd number in every class, first make the move on lamps $0, 1, 2$. The lamps that still differ from the target now form an even number in every class, and we finish as before. So exactly 1024 patterns are reachable.

Answer: 1024

Problem 3

A word is made of the letters A and B. A move replaces some occurrence of AB (an A immediately followed by a B) by BAA. Prove that, starting from any word, one cannot make moves for ever, and that the number of moves made before no move is possible does not depend on the choices.

For each letter A in the word, let k be the number of letters B to its right, and give that A the weight 2^k. Let W be the sum of the weights of all the As.

We claim W is invariant. Consider a move replacing AB by BAA, and suppose the A in the block has k Bs to its right, where $k \geq 1$ because of the B right after the A. After the move, the block reads BAA, and each of the two new As has $k-1$ Bs to its right, so their total weight is $2 \cdot 2^{k-1} = 2^k$, the same as before. Every A to the left of the block still has the same number of Bs to its right (the B has only moved within the block), and every A to the right of the block is unaffected. So W does not change.

Each move increases the number of As by exactly one. On the other hand every weight is at least 1, so the number of As is always at most W. If a_0 is the number of As at the start, at most $W - a_0$ moves can be made, so the process cannot go on for ever.

When no move is possible, the word contains no AB, so the word has the form B . . . BA . . . A. Then every A has weight $2^0 = 1$, and the number of As equals W. Since the number of As went up by one per move, the number of moves made is exactly $W - a_0$, which depends only on the starting word.

Problem 4

Ten coins lie in a row, all showing heads. In one move, choose a coin showing heads, turn that coin over, and also turn over the coin immediately to its right, if there is one. Prove that the moves must eventually stop, and find the largest possible number of moves.

Number the coins 1 to 10 from left to right and give coin i the weight $11 - i$. Let W be the total weight of the coins showing heads, and at the start $W = 10 + 9 + \cdots + 1 = 55$.

Consider a move on coin i, which shows heads. If $i = 10$, only coin 10 turns over and W drops by 1. If $i \leq 9$, coin i turns to tails, losing $11 - i$, and coin $i+1$ turns over. If coin $i+1$ was tails, W changes by $-(11-i) + (10-i) = -1$. If this coin was heads, W changes by $-(11-i) - (10-i) = -(21-2i) \leq -3$. So every move decreases W by at least 1. Since W is a non-negative integer, at most 55 moves can be made, and in particular the moves must stop.

To make 55 moves, always choose the rightmost coin showing heads. The coin to its right, if there is one, shows tails, so by the calculation above W drops by exactly 1. This can be repeated as long as some coin shows heads, that is, as long as $W > 0$. So we make exactly 55 moves, and the answer is 55.

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Answer: 55

Problem 5

In a chess tournament with $n \geq 3$ players, every two players played each other exactly once and no game was drawn. Every player won at least one game. Prove that there are three players A, B, C such that A beat B, B beat C and C beat A.

Call a sequence of distinct players $Q_1, Q_2, \ldots, Q_k$ with $k \geq 2$ a *cycle* if Q_1 beat Q_2, Q_2 beat Q_3, and so on, and Q_k beat Q_1.

First, a cycle exists. Start with any player P_1. Since every player won a game, P_1 beat some P_2, who beat some P_3, and so on. There are finitely many players, so some player appears twice, so take the first repetition $P_j = P_i$ with $i < j$. Then $P_i, P_{i+1}, \ldots, P_{j-1}$ are distinct and form a cycle. Its length is not 1, since nobody plays themselves.

Now choose a cycle $Q_1, \ldots, Q_k$ of smallest possible length. Two players play only once, so $k = 2$ is impossible, and $k \geq 3$. Suppose $k \geq 4$ and look at the game between Q_1 and Q_3. If Q_3 beat Q_1, then Q_1, Q_2, Q_3 is a cycle of length $3 < k$. If Q_1 beat Q_3, then $Q_1, Q_3, Q_4, \ldots, Q_k$ is a cycle of length $k - 1$. Either way we contradict the minimality of k. Hence $k = 3$, and $A = Q_1$, $B = Q_2$, $C = Q_3$ are the required players.

Problem 6

Six bowls stand in a circle. One of them, bowl A, contains a single bean, and the others are empty. In one move, take a bean out of any bowl and put one bean into each of the two neighbouring bowls. Prove that the bowl opposite A can never contain more beans than the other five bowls together.

Number the bowls $0, 1, \ldots, 5$ around the circle, with A as bowl 0, so that bowl 3 is opposite A, and indices are read modulo 6. Let c_i be the number of beans in bowl i.

A move at bowl k changes a weighted sum $\sum w_i c_i$ by $w_{k-1} + w_{k+1} - w_k$, so we want weights with $w_k = w_{k-1} + w_{k+1}$ for every k. Taking $w = (1, 0, -1, -1, 0, 1)$ works, as one checks $w_5 + w_1 = 1 = w_0$, $w_0 + w_2 = 0 = w_1$, $w_1 + w_3 = -1 = w_2$, $w_2 + w_4 = -1 = w_3$, $w_3 + w_5 = 0 = w_4$ and $w_4 + w_0 = 1 = w_5$. Hence

$$I = c_0 - c_2 - c_3 + c_5$$

is unchanged by every move. At the start $I = 1$, so at every later stage

$$c_3 = c_0 + c_5 - c_2 - 1 \leq c_0 + c_5 - 1 < c_0 + c_1 + c_2 + c_4 + c_5.$$

So bowl 3 always contains fewer beans than the other five bowls together.

(The weights were found by solving $w_{k+1} = w_k - w_{k-1}$. Any starting pair $1, x$ gives the sequence $1, x, x-1, -1, -x, 1-x$, which repeats with period 6, so the sequence fits round the circle. The choice $x = 0$ gives the simplest inequality.)

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